

Design and Characterization of Immune-Stimulating Imidazo[4,5-c]quinoline Antibody-Drug Conjugates

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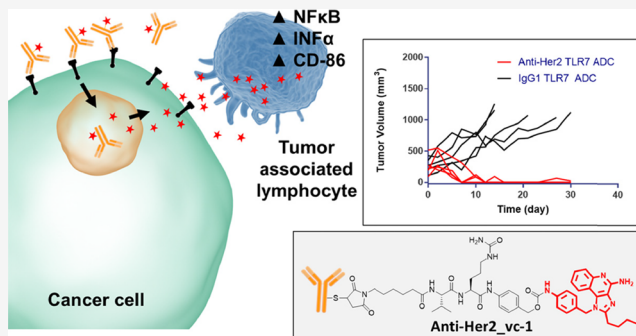


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ABSTRACT: Traditional antibody-drug conjugate (ADC) technology has employed tumor-targeting antibodies to selectively deliver ultrapotent cytotoxins to tumor tissue. While this technology has been highly successful, resulting in the FDA approval of over 10 ADCs, the field continues to struggle with modest efficacy and significant off-target toxicity. Concurrent with the struggles of the ADC field, a new generation of immune-activating therapeutics has arisen, most clearly exemplified by the PD-1/PD-L1 inhibitors that are now part of standard-of-care treatment regimens for a variety of cancers. The success of these immuno-oncology therapeutic agents has prompted the investigation of a variety of new immuno-stimulant approaches, including toll-like receptor (TLR) activators. Herein, we describe the optimization of ADC technology for the selective delivery of a potent series of TLR7 agonists. A series of imidazole[4,5-c]quinoline agonists (as exemplified by compound 1) were shown to selectively agonize the human and mouse TLR7 receptor at low nanomolar concentrations, resulting in the release of IFN α from human peripheral blood mononuclear cells (hPBMCs) and the upregulation of CD86 on antigen-presenting cells. Compound 1 was attached to a deglycosylated (Fc- γ null) HER2-targeting antibody via a cleavable linker, resulting in an ADC (**anti-HER2_vc-1**) that potently and selectively activated the TLR7 pathway in tumor-associated macrophages via a “bystander” mechanism. We demonstrated that this ADC rapidly released the TLR7 agonist into the media when incubated with HER2+ cells. This release was not observed upon incubation with an isotype control ADC and furthermore was suppressed by co-administration of the naked antibody. In co-culture experiments with HER2+ HCC1954 cells, this ADC induced the activation of the NF κ B pathway in mouse macrophages and the release of IFN α from hPBMCs, while a corresponding isotype control ADC did not. Finally, we demonstrated that IP administration of **anti-HER2_vc-1** induced complete tumor regression in an HCC1954 xenograft study in SCID beige mice. Unlike related ADC technology that has been reported recently, our technology relies on the passive diffusion of the TLR7 agonist into tumor-associated macrophages rather than Fc- γ -mediated uptake. Based on these observations, we believe that this ADC technology holds significant potential for both oncology and infectious disease applications.



KEYWORDS: antibody-drug conjugate, TLR7, cancer, immune stimulation, immuno-oncology

INTRODUCTION

Checkpoint inhibitors have revolutionized the treatment of cancer by promoting a sustained T-cell response against tumors, thereby promoting long-term tumor regression. These agents, such as programmed cell death protein 1/programmed death-ligand 1 (PD-1/PD-L1) inhibitors and cytotoxic T-lymphocyte-associated protein 4 (CTLA-4) inhibitors, are now part of “standard of care” treatment regimens for many types of cancer including melanoma, non-small cell lung cancer (NSCLC), and metastatic renal cell carcinoma (mRCC). Unfortunately, response rates to checkpoint inhibitors are typically only 15–30% for most solid tumors.¹ For this reason, there is rapidly growing interest in other immune-activating technology that may complement checkpoint therapies. Toll-

like receptor (TLR) agonists are one such class of molecules.^{2,3} TLRs are a class of pattern recognition receptors that play a critical role in the innate immune response, typically recognizing “danger-associated molecular patterns” (DAMPs) that are connected with viral and bacterial infections. TLRs signal through a MyD88/IRAK4 pathway that ultimately activates the nuclear factor kappa-light-chain-enhancer of

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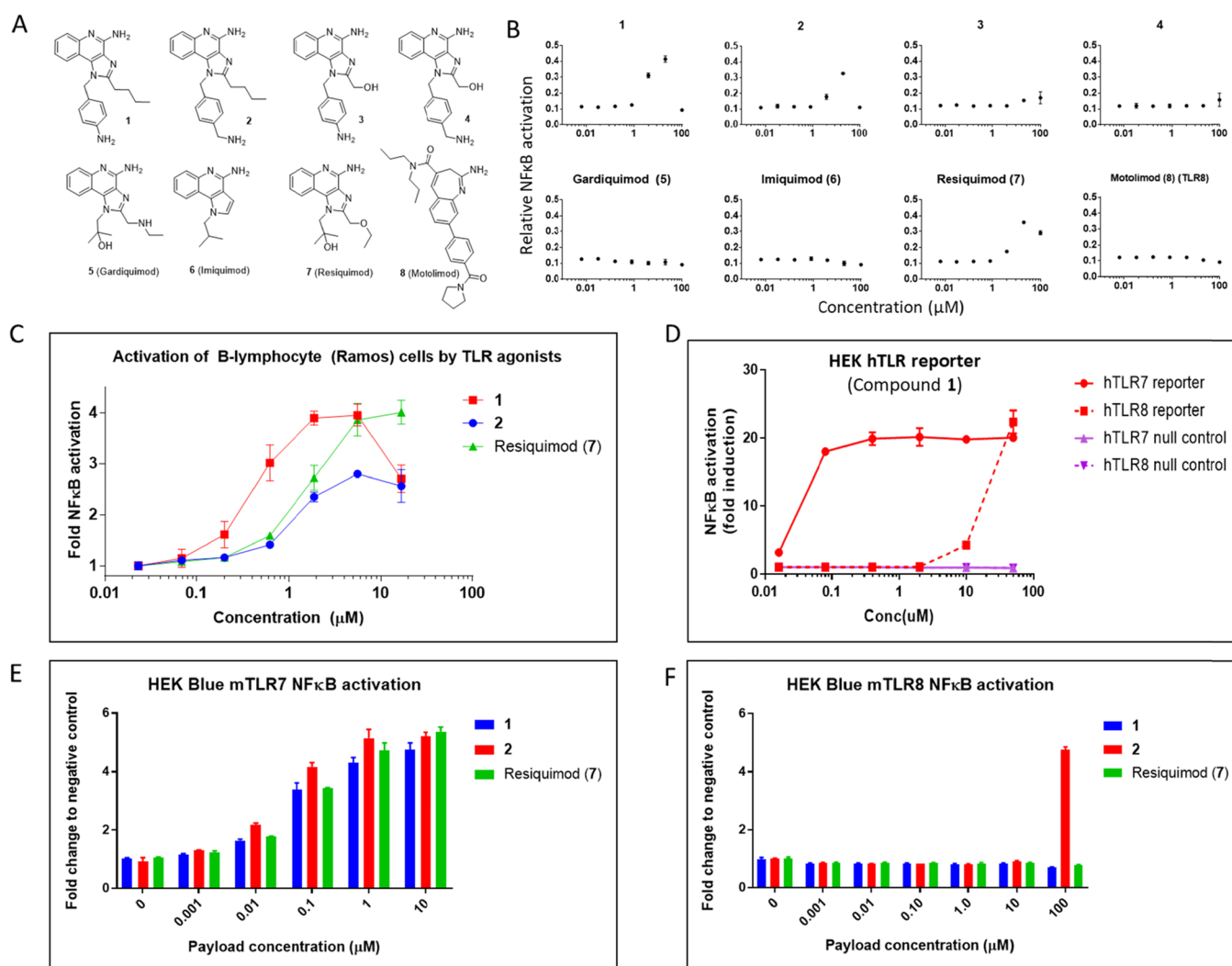


Figure 1. (A) Structures of TLR agonists screened; (B) evaluation of potential TLR agonists in a Ramos-blue NFκB reporter assay; (C) a 72 h activation assay in Ramos-blue cells; (D) activity of 1 in hTLR7 and hTLR8 reporter assays; (E,F) activity of 1, 2, and 7 in mTLR7 and mTLR8 reporter assays in HEK293 cells.

activated B cells (NFκB) and interferon regulatory factor (IRF) which, in turn, promote the release of inflammatory cytokines and type I interferons, respectively. Overactivation of TLR signaling pathways is known to play a role in the pathophysiology of various autoimmune disorders, including rheumatoid arthritis and lupus, thus illustrating the importance of this pathway in regulating adaptive immunity.^{2,4} In light of this, a variety of agents have been developed to block downstream TLR signaling in order to dampen the impact of TLR activation.^{5,6} Conversely, the activation of various TLR pathways has been broadly explored as a means of immune stimulation for cancer immunotherapy and for the treatment of viral infections.^{3,7}

There are 10 subtypes of TLR receptors in humans, six of which are found on the plasma membrane (TLRs 1, 2, 4–6, 10) and four of which are found on the endosomal membrane (TLRs 3, 7–9). Membrane-bound TLRs are typically activated by bacterial- and viral-derived proteins and lipids, while the endosomal TLRs are primarily activated by viral and bacterial nucleic acids.⁸ Specifically, TLR7 and TLR8 recognize viral-derived ssRNA, resulting in the production of Th1-polarizing cytokines and interferons, including TNFα, IL-12, and IFNα. TLR7 is primarily expressed in plasmacytoid dendritic cells

(pDCs), B-cells, and, to a lesser extent, myeloid cells.⁹ TLR8 is typically not expressed in pDCs and B cells but is highly expressed by myeloid cells such as monocytes, macrophages, and myeloid dendritic cells.⁹ While TLR7 and TLR8 have different expression profiles and downstream effects, they often have overlapping ligand-binding domains and thus pharmacology—sharing many of the same structural features in their known agonists. Indeed, a variety of potent dual TLR7/8 agonists have been reported over the past 10 years, being widely explored for their potential as vaccine adjuvants, antiviral agents, and anti-cancer agents.¹⁰

Given the role of TLR7 and TLR8 in the recognition of nucleic acids, it is not surprising that many known agonists of these receptors have purine-like features, including fused 6,5 nitrogen-containing ring systems with pendant amine functionalities. For example, imiquimod (an imidazo[4,5-c]quinoline) is a potent TLR7/8 agonist that is widely used as a topical antiviral agent by virtue of its ability to activate the innate and adaptive immune system via interferon-α (IFNα) release.¹¹ Several structurally similar analogues, including gardiquimod and resiquimod, have been reported in literature and evaluated in the clinic (Figure 1A). While these TLR agonists were initially designed for antiviral and vaccine

adjuvant applications, there has been rapidly increasing interest in the role that TLR7/8 may play in tumor immunogenicity. Tumor-associated macrophages (TAMs) are typically found to be in an immune-suppressive (M2-like) phenotype, perhaps due to the action of regulatory T-cells (Tregs) and their associated release of IL-10. Various studies have shown that TLR7/8 agonists, such as resiquimod, can transform M2-like macrophages to immunologically active M1-like macrophages which support tumor regression both as single agents and in synergy with checkpoint inhibitors.^{12,13} Likewise, TLR7/8 agonists have been shown to activate tumor-associated myeloid dendritic cells and pDCs, both important for the recruitment and activation of T-cells.^{13,14} Indeed, the TLR7/8 agonist imiquimod is topically used for the treatment of basal cell carcinoma and melanoma.¹⁵ Unfortunately, systemic administration of TLR7/8 agonists has been associated with severe toxicity due to cytokine release syndrome and related inflammatory adverse reactions.¹⁶ For this reason, there has been an intense effort to develop technologies for tumor-specific delivery of TLR7/8 agonists, including intratumoral injections, nanoparticle formulations, prodrugs cleaved in the tumor microenvironment, and (most recently) antibody conjugates.^{13,17–21} Herein, we report the design and optimization of a breast-cancer-targeting antibody-drug conjugate (ADC) that delivers a potent TLR7 agonist directly to lymphocytes.

ADC technology is a tissue-targeted delivery approach that is broadly useful for improving the therapeutic window of highly potent pharmaceutical agents.²² This technology has resulted in the FDA approval of 11 marketed drugs and innumerable clinical-stage programs. ADCs are designed to deliver a small-molecule therapeutic agent to a select tumor tissue type by virtue of linking it to an antibody that binds to a selectively expressed tumor antigen. ADCs are composed of a monoclonal antibody (mAb), a chemical linker, and a therapeutic payload, each of which plays a critical role in the efficacy and pharmacokinetics of the ADC. Selective payload delivery to the targeted tissue is achieved through judicious selection of a mAb that exhibits high affinity toward a selectively expressed cell surface antigen. The ADC then undergoes clathrin-mediated endocytosis, resulting in transport to lysosomal compartments that, in turn, promote ADC catabolism and payload release. ADCs that contain so-called “cleavable linkers” release their cargo upon processing by lysosomal enzymes such as proteases and glycosidases. On the other hand, ADCs containing “non-cleavable linkers” undergo full proteolysis of the antibody backbone, releasing a modified payload species that still contains vestigial components of the linker residue. All marketed ADCs to date deliver cytotoxic payloads such as tubulin inhibitors, DNA crosslinkers, and topoisomerase inhibitors. However, as described above, the success of checkpoint inhibitors has resulted in significant interest in immune-stimulating ADC technology. Due to our background and interest in ADC design,^{23–26} we undertook an effort to prepare and evaluate ADCs that would deliver TLR7 agonists to tumor tissues. In contrast to recently reported immune-activating ADC technology,^{17,20,21} the focus of our efforts was to develop tumor-targeted bioconjugates with cleavable linkers and cell-permeable payloads that could broadly activate TAMs and dendritic cells in the tumor microenvironment.

With this in mind, we selected a panel of 8 compounds (Figure 1) for screening based on various literature reports of

TLR7/8 agonists. The compounds were initially assessed in TLR-reporter assays and were later characterized in cytokine release and flow cytometry studies to verify their utility in immune-activating ADC technology. Compound 1 emerged from these studies as a highly potent and selective TLR7 agonist that was used for the design of various ADCs. Herein, we describe the identification, preparation, and evaluation of ADCs derived from 1, 2, and resiquimod in both in vitro and in vivo tumor models.

EXPERIMENTAL SECTION

General Experimental Information. All chemical reagents were purchased commercially and were used without further purification. ¹H NMR was performed using a Bruker Avance III HD 400 MHz. The identity and purity of all compounds were determined by ¹H NMR and LCMS (see methodology below). Antibodies were obtained as follows: anti-HER2/trastuzumab (SydLabs), anti-VEGFR2/ramucirumab (SydLabs), anti-IL6R/tocilizumab (SydLabs), anti-CD38/daratumumab (SydLabs), anti-CD20/rituximab (SydLabs), anti-79b/polatuzumab (Genentech), anti-TNF α /adalimumab (SydLabs), and anti-CD11a/efaluzumab (Absolute Antibody). Fluorophore-labeled antibodies were obtained from ThermoFisher or VWR: anti-CD14-PE (VWR 561707), anti-CD40-APC (Fisher 50-112-3049), anti-CD20-PE-Cy7 (Fisher 25-0209-41), and anti-CD86-AF488 (Fisher 50-167-09). Mouse TLR7 and TLR8-expressing HEK-Blue cells were purchased from InvivoGen (cat# hkb-mtlr7 and hkb-mtlr8).

Preparation of 1-(4-Aminobenzyl)-2-butyl-1H-imidazo[4,5-c]quinolin-4-amine (1). *Step 1. tert-Butyl (4-(((3-Nitroquinolin-4-yl)amino)methyl)phenyl)carbamate (11a).* To a solution of 4-chloro-3-nitroquinoline (9) (2.02 g, 1.0 equiv, 9.70 mmol) in DCM (30.0 mL) were added *tert*-butyl (4-(aminomethyl)phenyl)carbamate (10a) (2.15 g, 1.0 equiv, 0.970 mmol) and triethylamine (1.45 g, 2.0 mL, 1.5 equiv, 14.3 mmol). The mixture was refluxed at 40 °C for 1 h. The reaction progress was monitored by UPLC. Complete conversion of the reactants to the desired product was achieved by 1 h, forming a yellow precipitate. The reaction mixture was cooled to rt, concentrated to dryness, suspended into water (100 mL), filtered under vacuum, and dried through a desiccator to obtain the desired compound (3.35 g, 87.5% yield) as a bright yellow solid. HPLC *rt* = 3.34 min (polar gradient); *m/z* = 395.3 [M + H]. ¹H NMR (400 MHz, DMSO) δ_{H} /ppm 9.41 (s, 1H), 9.36 (s, 1H), 9.02 (s, 1H), 8.68 (d, *J* = 8.2 Hz, 1H), 7.90 (d, *J* = 8.3 Hz, 1H), 7.83 (t, *J* = 8.1 Hz, 1H), 7.57 (t, *J* = 8.3 Hz, 1H), 7.41 (d, *J* = 8.5 Hz, 2H), 7.22 (d, *J* = 8.5 Hz, 2H), 4.83 (s, 2H), 1.46 (s, 9H).

Step 2. tert-Butyl (4-(((3-Aminoquinolin-4-yl)amino)methyl)phenyl)carbamate (12a). To a suspension of the (*tert*-butyl (4-(((3-nitroquinolin-4-yl)amino)methyl)phenyl)carbamate (3.35 g, 1.0 equiv, 8.48 mmol) in MeOH (40.0 mL) were added zinc (2.74 g, 5.0 equiv, 42.5 mmol) and ammonium chloride (2.26 g, 5.0 equiv, 42.5 mmol). The reaction mixture was stirred at 40 °C for 10 min (to give a gray suspension) and monitored by UPLC. Then, the product formed immediately. After 10 min, the reaction mixture was filtered through celite, and the solvent was evaporated in vacuo. The residue was dissolved in DCM and partitioned against water and then extracted with DCM (2 $\times$ 100 mL). The organic layer was filtered through celite and dried over MgSO₄. The solvent was removed under vacuum to obtain the desired compound (2.96 g, 95.7%) as a golden-amber, web-like

crystal. HPLC *rt* = 2.66 (polar gradient); *m/z* = 365.3 [*M* + *H*].

Step 3. *tert*-Butyl 4-((2-Butyl-1*H*-imidazo[4,5-*c*]quinolin-1-yl)methyl)phenyl)carbamate (13a). To the crude product of step 2 (*tert*-butyl 4-(((3-aminoquinolin-4-yl)amino)-methyl)phenyl)carbamate, 2.96 g, 1.0 equiv, 8.12 mmol) in anhydrous EtOAc (75.0 mL), cooled to 0 °C, was added previously cooled triethylamine (1.07 g, 1470 μ L, 1.3 equiv, 10.6 mmol). The reaction was stirred at *rt* for 10 min. Thereafter, valeryl chloride (1.08 g, 1060 μ L, 1.1 equiv, 8.93 mmol) in EtOAc (25.0 mL) was added dropwise at 0 °C, and the reaction mixture was further stirred for 15 min and monitored by UPLC. The solution was concentrated to dryness under reduced pressure and carried forward without an intermediate purification step. LCMS *rt* = 2.88 min (polar gradient); *m/z* = 449.3 [*M* + *H*]. The crude product (3.64 g, 8.12 mmol) was dissolved in EtOH (49.0 mL) and treated with NaOH (644 mg, 2.0 equiv, 16.2 mmol) in H₂O (7.5 mL). The reaction mixture was refluxed at 80 °C for 2 h, and progress was monitored by UPLC. Upon completion, the solution was added to 100 mL of water, and this was washed with DCM (2 $\times$ 100 mL). The organic layer was dried over MgSO₄ and evaporated to dryness. The dried amber-brown resin (3.35 g) was purified through silica gel chromatography using 5% MeOH in DCM. The resulting eluent was concentrated to dryness, affording the desired compound (1.940, 55.4%) as an auburn crystal. LCMS *rt* = 3.13 (polar gradient); *m/z* = 431.3 [*M* + *H*]. ¹H NMR (400 MHz, DMSO) δ_{H} /ppm 9.31 (s, 1H), 9.20 (s, 1H), 8.11 (d, *J* = 7.8 Hz, 2H), 7.60 (t, *J* = 8.3 Hz, 1H), 7.49 (t, *J* = 8.3 Hz, 1H), 7.38 (d, *J* = 8.6 Hz, 2H), 6.93 (d, *J* = 8.6 Hz, 2H), 5.87 (s, 2H), 2.96 (t, *J* = 7.6 Hz, 2H), 1.78 (quint, *J* = 7.5 Hz, 2H), 1.44 (m, *J* = 7.7, 11H), 0.89 (t, *J* = 7.4 Hz, 3H).

Step 4. *tert*-Butyl 4-((4-Amino-2-butyl-1*H*-imidazo[4,5-*c*]quinolin-1-yl)methyl)phenyl)carbamate. *tert*-Butyl 4-((2-butyl-1*H*-imidazo[4,5-*c*]quinolin-1-yl)methyl)phenyl)carbamate (1.78 g, 1.0 equiv, 4.12 mmol) in CHCl₃ (25.0 mL) was treated with 3-chlorobenzoperoxoic acid (1.53 g, 1.50 equiv, 6.20 mmol) and stirred at 40 °C for 1 h. Ammonium hydroxide (28–38%) (22.0 g, 25.0 mL, 30.5 equiv, 126 mmol) was added dropwise to the reaction and then left stirring for 1 h. After this, 4-methylbenzenesulfonyl chloride (1.58 g, 2.0 equiv, 8.24 mmol) was added at once, and the mixture was stirred at room temperature for 1 h and monitored by UPLC. Upon completion, the reaction was diluted with 75.0 mL of CHCl₃, and the organic fraction was washed with 75 mL of aqueous NaHCO₃ and then washed with brine (100 mL). The organic layer was dried over MgSO₄, and the solvent was removed under reduced pressure to obtain the crude product (2.38 g, 130%) which was used for the next step without further purification. LCMS *rt* = 2.92 min; *m/z* = 446.3 [*M* + *H*].

Step 5. 1-(4-Aminobenzyl)-2-butyl-1*H*-imidazo[4,5-*c*]quinolin-4-amine (1). To a solution of crude extract of *tert*-butyl 4-((4-amino-2-butyl-1*H*-imidazo[4,5-*c*]quinolin-1-yl)methyl)phenyl)carbamate (1.84 g, 1.0 equiv, 4.12 mmol) in DCM (45.0 mL) was added TFA (5.0 mL). The mixture was stirred at *rt* for 3 h, and the solvent was removed in vacuo. The resulting black oil was purified by silica gel chromatography using 10% MeOH in DCM. The eluent was concentrated to dryness, affording the desired material (1.07 g, 75% from 13a) as a yellow-amber crystal. LCMS: *rt* = 2.52 min (HPLC gradient). *m/z* 345.3 [*M* + *H*]⁺; ¹H NMR (400 MHz, DMSO)

δ_{H} /ppm 9.13 (s, 2H), 7.98 (d, *J* = 7.8 Hz, 1H), 7.77 (d, *J* = 8.0 Hz, 1H), 7.63 (t, *J* = 8.3, 1H), 7.37 (t, *J* = 8.2 Hz, 1H), 6.94 (d, *J* = 8.5 Hz, 2H), 6.86 (d, *J* = 8.4 Hz, 2H), 5.85 (s, 2H), 2.96 (t, *J* = 7.7 Hz, 2H), 1.72 (quint, *J* = 7.6 Hz, 2H), 1.39 (sextet, 7.4 Hz, 2H), 0.873 (t, *J* = 7.3 Hz, 3H).

Preparation of 1-(4-(Aminomethyl)benzyl)-2-butyl-1*H*-imidazo[4,5-*c*]quinolin-4-amine (2). Compound 2 was prepared using slight modifications of the methodology outlined for compound 1 above.

LC–MS: *m/z* 359.48 [*M* + *H*]⁺; retention time = 1.99 min. ¹H NMR (400 MHz, DMSO) δ_{H} /ppm 11.57 (s, 2H), 8.14 (dd, *J* = 5.5, 2.6 Hz, 1H), 7.43 (s, 2H), 7.41 (d, *J* = 1.9 Hz, 2H), 7.39 (s, 1H), 7.37–7.31 (m, 2H), 7.10 (dd, *J* = 18.7, 8.3 Hz, 2H), 5.85 (s, 2H), 3.99 (dt, *J* = 11.2, 5.7 Hz, 2H), 1.72 (dt, *J* = 15.3, 8.2 Hz, 2H), 1.41–1.34 (m, 2H), 0.90–0.85 (m, 3H).

Preparation of 4-(2-(2-(6-(2,5-Dioxo-2,5-dihydro-1*H*-pyrrol-1-yl)hexanamido)-3-methylbutanamido)-5-ureidopentanamido)benzyl 4-((4-Amino-2-butyl-1*H*-imidazo[4,5-*c*]quinolin-1-yl)methyl)phenylcarbamate [mcValCitPABC-1]. To a solution of 1-(4-aminobenzyl)-2-butyl-1*H*-imidazo[4,5-*c*]quinolin-4-amine (1) (9.1 mg, 26 μ mol) in DMA (500 μ L) were added mcValCit-PAB-PNP (23 mg, 32 μ mol), 2,6-dimethylpyridine (9.2 μ L, 79 μ mol), and HOBt (5.6 mg, 37 μ mol). The mixture was stirred overnight at *rt*. The crude mixture was purified by prep-HPLC to give an amber-colored sticky solid (36.2 mg, 60.3%). LC–MS: *m/z* 944.1 [*M* + *H*]⁺; retention time = 3.27 min. ¹H NMR (400 MHz, DMSO) δ_{H} /ppm 9.97 (s, 1H), 7.79 (d, *J* = 1.2 Hz, 1H), 7.59 (d, *J* = 8.8 Hz, 1H), 7.40 (d, *J* = 9.2 Hz, 1H), 7.32 (d, *J* = 8.7 Hz, 1H), 7.00 (s, 1H), 6.99 (s, 2H), 6.97 (d, *J* = 6.6 Hz, 1H), 6.51 (s, 2H), 5.39 (s, 1H), 5.04 (s, 1H), 2.77 (d, *J* = 2.7 Hz, 2H), 2.69–2.65 (m, 11H), 2.33 (dt, *J* = 3.9, 2.1 Hz, 9H), 2.08 (s, 3H), 1.95 (s, 4H), 1.24 (s, 3H), 0.89–0.86 (m, 2H), 0.85 (t, *J* = 1.8 Hz, 2H), 0.84–0.80 (m, 2H).

Preparation of 4-(2-(2-(6-(2,5-Dioxo-2,5-dihydro-1*H*-pyrrol-1-yl)hexanamido)-3-methylbutanamido)-5-ureidopentanamido)benzyl 4-((4-aminobenzyl)-2-butyl-1*H*-imidazo[4,5-*c*]quinolin-4-amine (2). 2,6-Dimethylpyridine (7.9 μ L, 68 μ mol) was added to a solution of 1-(4-(aminomethyl)benzyl)-2-butyl-1*H*-imidazo[4,5-*c*]quinolin-4-amine (2) (8.2 mg, 23 μ mol) in DMA (400 μ L), and the mixture was stirred for 15 min at 25 °C. HOBt (4.2 mg, 1.2 equiv, 27 μ mol) and mcValCitPAB-PNP (14 mg, 25 μ mol) in DMA (200 μ L) were added, and the reaction mixture was stirred overnight at *rt*. A saturated solution of NaHCO₃ (3 mL) was added, and the product was extracted with CH₂Cl₂ (3 $\times$ 5 mL). The organic fraction was dried over anhydrous MgSO₄, and the solvent was evaporated in vacuo. The crude extract was dissolved in DMA (500 μ L) and purified by preparative HPLC to yield a white solid (16.4 mg, 75%). LC–MS: *m/z* 958.5 [*M* + *H*]⁺; retention time = 2.55 min. ¹H NMR (400 MHz, DMSO) δ_{H} /ppm 10.06 (s, 1H), 9.92 (dd, *J* = 76.6, 35.4 Hz, 2H), 8.07 (dd, *J* = 14.4, 7.5 Hz, 3H), 8.02 (d, *J* = 8.4 Hz, 1H), 7.81 (d, *J* = 8.6 Hz, 4H), 7.61 (d, *J* = 8.5 Hz, 5H), 7.58–7.50 (m, 4H), 7.45–7.35 (m, 5H), 7.32 (d, *J* = 8.5 Hz, 2H), 7.23 (d, *J* = 8.5 Hz, 1H), 7.00 (s, 4H), 6.32–5.59 (m, 22H), 5.53 (s, 2H), 5.06 (d, *J* = 10.3 Hz, 2H), 4.43 (s, 1H), 4.42–4.33 (m, 3H), 4.19 (t, *J* = 7.7 Hz, 3H), 3.37 (t, *J* = 7.0 Hz, 6H), 3.08–2.90 (m, 7H), 2.25–2.06 (m, 7H), 2.06–1.86 (m, 3H), 1.71 (dt, *J* = 14.4, 7.2 Hz, 4H), 1.59 (dd, *J* = 13.5, 4.3 Hz, 3H), 1.48 (dt, *J* = 14.7, 7.3 Hz, 14H), 1.37 (dd, *J* = 14.9, 7.4 Hz, 5H), 1.24–1.13 (m, 6H), 0.84 (dd, *J* = 12.4, 6.7 Hz, 19H).

Preparation of 4-(2-(2-(6-(2,5-Dioxo-2,5-dihydro-1H-pyrol-1-yl)hexanamido)-3-methylbutanamido)-5-ureidopentanamido)benzyl-2-(ethoxymethyl)-1-(2-hydroxy-2-methylpropyl)-1H-imidazo[4,5-c]quinolin-4-ylcarbamate (mcValCitPABC-7). To a solution of resiquimod (7) (20 mg, 64 μ mol) in DMF (300 μ L) was added 2,6-dimethylpyridine (22 μ L, 0.19 mmol). After stirring at rt for 15 min, HOBt (12 mg, 76 μ mol) and mcValCitPABC-PNP (40 mg, 70 μ mol) in DMF (300 μ L) were added. After stirring overnight, water (5 mL) was added, and the product was extracted into EtOAc (3 $\times$ 5 mL). The organic layer was dried over anhydrous MgSO_4 and concentrated to dryness. The crude extract was dissolved in DMA (500 μ L) and purified by a reverse phase preparative HPLC (10% $\rightarrow$ 90% acetonitrile in water over 10 min, each solvent containing 0.05% TFA) to produce 6.5 mg (11.2%) of the title compound as a white solid. LC-MS: m/z 913.5 [$\text{M} + \text{H}$] $^+$; retention time = 2.88 min. ^1H NMR (400 MHz, DMSO) δ_{H} /ppm 10.02 (s, 1H), 8.65 (d, J = 8.4 Hz, 1H), 8.18–8.02 (m, 2H), 7.80 (d, J = 8.7 Hz, 2H), 7.65 (d, J = 8.6 Hz, 2H), 7.44 (d, J = 8.6 Hz, 1H), 7.00 (s, 2H), 5.25 (s, 1H), 4.39 (dd, J = 16.2, 9.1 Hz, 2H), 4.24–4.13 (m, 1H), 3.56 (q, J = 7.0 Hz, 2H), 3.37 (t, J = 7.0 Hz, 2H), 2.95 (s, 4H), 2.79 (s, 5H), 2.20–2.06 (m, 3H), 1.96 (s, 6H), 1.48 (dt, J = 15.0, 7.3 Hz, 6H), 1.18 (dt, J = 14.0, 7.6 Hz, 10H), 0.84 (dd, J = 12.2, 6.8 Hz, 7H).

Synthesis of Antibody-Drug Conjugations. The linker-payloads described were conjugated with various antibodies, resulting in the ADCs shown in Table 1. Note that all

Table 1. Preparation and Characterization of Immune-Activating ADCs

antibody	linker-payload	DAR	actual mass shift	theoretical mass shift	aggregation (%)
anti-HER2	mcValCitPABC_1	8.7	944	944	8.1%
anti-HER2	mcValCitPABC_2	8.0	960	958	<5%
anti-HER2	mcValCitPABC_7	7.3	914	913	<5%
anti-CD20	mcValCitPABC_1	8.8	943	944	6.9%
anti-CD20	mcValCitPABC_2	8	958	958	<5%
anti-CD20	mcValCitPABC_7	7.3	913	913	<5%
anti-RSV	mcValCitPABC_1	7.3	944	944	<5%

antibodies were deglycosylated with PNGase F prior to conjugation in order to minimize Fc- γ -mediated uptake.²⁷ The antibody was diluted to 2 mg/mL in PBS containing 5 mM EDTA and subsequently treated with TCEP (7–8 eq). After 2 h at 37 $^{\circ}\text{C}$, 25 equiv of the appropriate linker-payload was added (5 mM stock solution in DMA) along with sufficient DMA to result in a final organic of $\sim$ 5% (vol/vol). After 90 min at rt, the reaction was buffer exchanged into 100% PBS using a PD minitrapp G25 column according to the manufacturer's protocol. The drug-antibody ratio (DAR) was determined by LCMS, and the amount of aggregation was established by size exclusion chromatography (SEC) (Figures S7–8). The final ADC was filter sterilized prior to storage. Some ADCs exhibited slight loading over the theoretical DAR of 8 presumably due to partial loading to buried disulfides or

due to small amounts of non-specific reaction of the maleimide with lysine residues. Endotoxin levels were assessed using the ToxinSensor Chromogenic LAL Endotoxin Assay Kit (GenScript) (Table S1).

NF- κ B Activation Assay in Ramos-Blue Cells. Each payload (as a stock solution in DMA) was diluted in a 5 $\times$ serial dilution in 10% DMSO in PBS. Ramos-blue cells (InvivoGen, cat# rms-sp) were cultured using high-glucose IMDM supplied with 10% FBS according to the manufacturer's guidelines. The medium was supplemented with 50 $\mu\text{g}/\text{mL}$ penicillin, 50 $\mu\text{g}/\text{mL}$ streptomycin, 100 $\mu\text{g}/\text{mL}$ normocin, and 500 $\mu\text{g}/\text{mL}$ zeomycin to prevent bacterial contamination. Cells were seeded (in triplicate or quadruplicate) at a density of 0.2×10^6 cells/mL per well. 135 μL of the cell suspension was added to each well in a 96-well plate along with 15 μL of the corresponding payload. Cells were incubated at 37 $^{\circ}\text{C}$ under 5% CO_2 for 24 or 72 h. The cells were centrifuged at 1990 rpm for 10 min, and 40 μL of the cell supernatant was added to 160 μL of the SEAP substrate (Quanti-Blue, Invivogen cat# rep-qbs). After incubation at 37 $^{\circ}\text{C}$ for 24 h, the absorbance at 630 nm was determined using a Molecular Devices i3 $\times$ plate reader.

Evaluation of Payloads in HEK-Blue mTLR7, hTLR7, mTLR8, and hTLR8 Cells. Cells were maintained in culture media using high-glucose DMEM supplied with 10% FBS, supplemented with 50 U/mL penicillin, 50 $\mu\text{g}/\text{mL}$ streptomycin, and 100 $\mu\text{g}/\text{mL}$ normocin to prevent bacterial contamination. Prior to the experiment, cells were rinsed and detached using prewarmed DPBS and collected by centrifugation at 1100 rpm for 5 min. Cells were then re-suspended at 0.2 million cells per milliliter and seeded into 96-well plates (90 μL). Compounds 1, 2, and 7 were diluted to 10 $\times$ the final desired concentrations in PBS. 10 μL of the payload solution (or PBS blank) was added to each well. Each experiment was performed in triplicate. The plate was incubated at 37 $^{\circ}\text{C}$ /5% CO_2 environment for 24 h. The supernatants were collected, and the SEAP induction (NF- κ B activation) was determined using a QuantiBlue substrate, per the manufacturer's protocol, by measuring the absorbance at 630 or 650 nm.

TNF- α Induction in THP-1 Monocytes and Macrophages. The human monocyte line THP-1 was purchased from ATCC and maintained in RPMI 1640 media supplied with 10% FBS. Differentiation into macrophages was accomplished as follows: THP-1 cells were seeded to 96-well plates (20,000 cells per well) and cultured in 100 μL of RPMI 1640 media supplied with 10% FBS containing 200 nM phorbol-12-myristate-13-acetate (PMA) for 72 h. After the incubation, the supernatant was removed, and the cells were rinsed with pre-warmed DPBS twice to eliminate the remaining PMA. The cells were then cultured in fresh RPMI 1640 media supplied with 10% FBS for at least 72 h. Morphological evidence of macrophage differentiation was confirmed microscopically. Cytokine release from THP-1 and THP-1-differentiated macrophages was accomplished as follows: THP-1 monocytes were seeded to 96-well plates at a density of 0.2 million cells per milliliter with a volume of 90 μL . THP-1 macrophages were supplied with 90 μL of fresh culture media. Stock solutions of the payloads were prepared at 10 $\times$ the final test concentration. 10 μL of the payload solutions was added to the corresponding wells and mixed thoroughly. Each experiment was performed in triplicate, and penicillin/streptomycin was used to avoid bacterial contamination. 10 μL of DPBS was added to the blank to keep volumes consistent. The cells were incubated in

37 °C/5% CO₂ for 24 h. After the incubation, the plates were centrifuged, and the supernatant samples were collected. TNF alpha levels in each supernatant sample were detected using a DuoSet TNF alpha ELISA kit (R&D systems). The data was analyzed and plotted using GraphPad Prism 7 software.

Cytokine Induction and CD86 Induction in PBMCs. A stimulation assay was performed to test if compounds 1, 2, and 7 could activate primary immune cells in vitro. In brief, human peripheral blood mononuclear cells (hPBMCs, iXCells Biotech) were thawed and seeded in triplicate to a 96-well plate at a seeding number of ~200,000 cells per well with a volume of 90 μ L. PBMCs were maintained in RPMI 1640 media supplied with 10% fetal bovine serum for 24 h before treatment. The appropriate compound was added in 10 μ L working solutions which were prepared at 10 $\times$ the final test concentration. Non-treated cells were used as blank. The cells were incubated under 37 °C/5%CO₂ for 24 h. The plates were spun down to separate cells from the supernatant. Cytokines including TNF alpha, IL-6, IFN gamma, and IFN alpha were analyzed via the ELISA assay (R&D systems, PBL scientific) following the manufacturer's protocol. The data was analyzed and plotted using GraphPad Prism 7 software. The cell pellets were collected into minicentrifuge tubes, rinsed with DPBS supplied with 1% BSA, and incubated with fluorophore-labeled antibodies (anti-CD14-PE, anti-CD40-APC, anti-CD20-PE-Cy7, and anti-CD86-AF488) according to the manufacturer's protocol. The antibody incubation was performed on ice to minimize internalization. The cells were then analyzed using a BD Accuri C6 plus flow cytometer. CD14 positive or CD20 positive cells were gated as monocytes/macrophages or B lymphocytes, respectively. CD86 expression levels were represented and plotted as MFI values of AF488.

Shake-Flask LogD. A 20 mM stock solution of the small-molecule payload (100 μ L, in DMSO) was added to 2 mL of 1:1 PBS:octanol. The mixture was shaken at room temperature at 1500 rpm overnight, filtered, and then centrifuged. A 50 μ L aliquot of the aqueous layer was added to 60 μ L of 1:6 ACN:H₂O for analysis (2.2 $\times$ dilution factor). A 10 μ L aliquot of the octanol layer was added to 250 μ L of DMSO. A 5 μ L aliquot of the resulting solution was added to 57.5 μ L of 1:6 ACN:H₂O (325 $\times$ dilution factor). The amount of payload in each layer was determined by LCMS/MS against a standard curve of the corresponding payload (1 μ M to 1 nM) injected in 1:6 ACN:H₂O. Each measurement was performed in triplicate.

LCMS Payload Release Assay with SKBR3 Cells. SK-BR-3 were seeded in triplicate to a 24-well plate with a seeding number of ~100,000 cells per well. The cells were cultured in 450 μ L of RPMI 1640 media supplied with 10% fetal bovine serum. 50 μ L of working solutions of targeted (anti-HER2) or non-targeted (anti-CD20) ADCs was added to the cells. Supernatant samples (100 μ L) at different time points were collected. The samples were treated with acetonitrile (300 μ L), and samples were centrifuged to remove proteins. The supernatant was evaporated and reconstituted with 100 μ L of 25% acetonitrile in water. The concentration of payload was then determined by LCMS/MS (360.2 $\rightarrow$ 119.95) against a freshly prepared standard curve.

Conditioned Media Assay with SKBR3 Cells. SKBR3 cells were maintained in high-glucose DMEM supplied with 10% fetal bovine serum. Cells were seeded to 96-well plates (100 μ L, 20 k cells) and allowed to adhere overnight. HEK-Blue mTLR7 cells were also seeded to 96-well plates as previously

described with a density of 0.2 million cells per milliliter and a volume of 90 μ L. TLR7 agonist ADCs were diluted to 10 $\times$ the final test concentration in PBS. 10 μ L of ADC solution, ADC solution with naked monoclonal antibody, and DPBS were added to the corresponding wells in triplicate, mixed thoroughly, and incubated in a 37 °C/5%CO₂ environment for 48 h. After the incubation, the supernatant of the HEK-Blue mTLR7 plates was replaced by the supernatants of the corresponding wells of the SKBR3 plates. The HEK-Blue mTLR7 cells were incubated for 48 h before the supernatants were collected and evaluated for the secretion of SEAP as described above. The data was analyzed and plotted using GraphPad Prism 7 software.

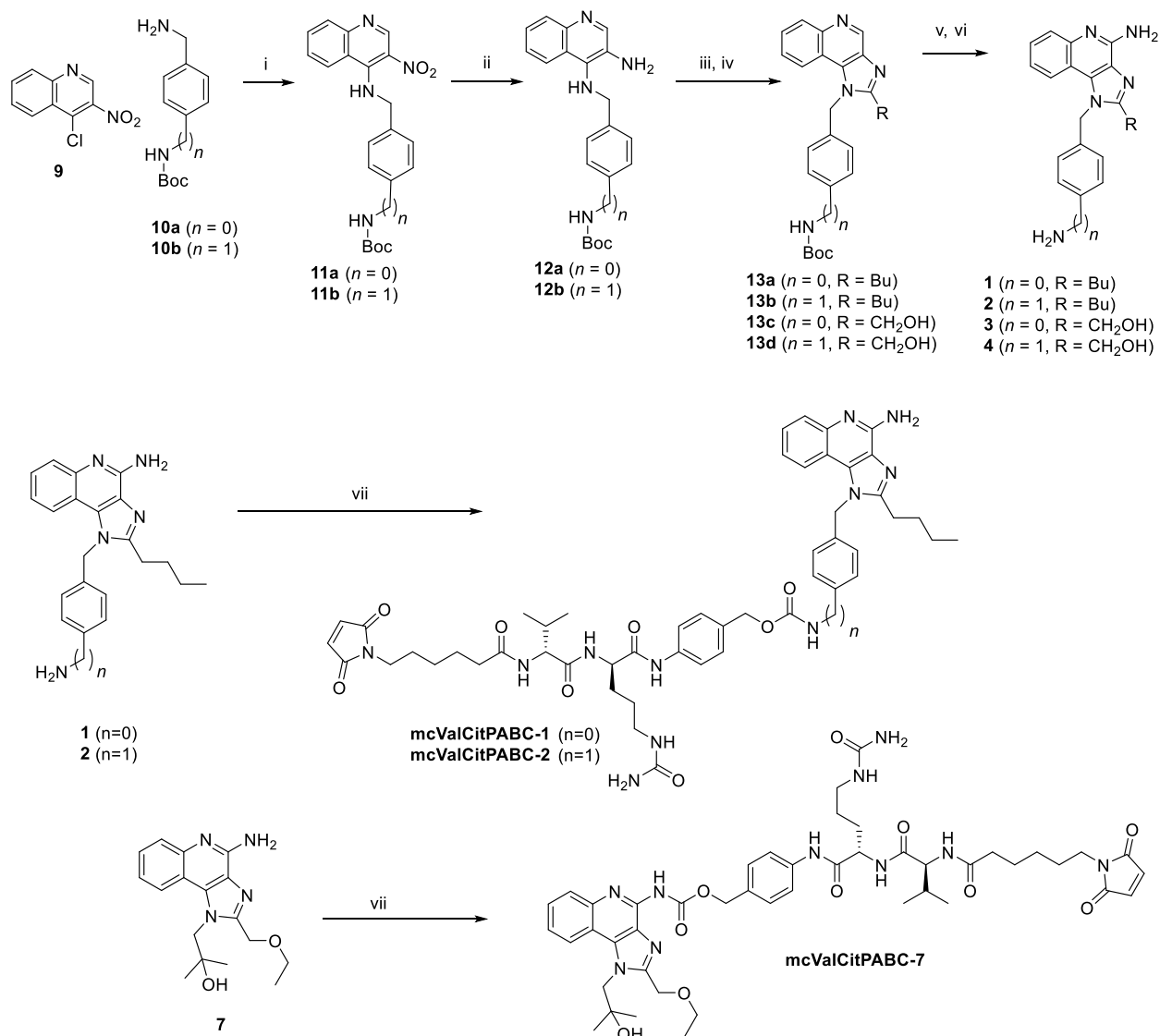
Cytotoxicity of ADCs against SKBR3 and HCC1954 Cells. SKBR3 and HCC1954 cells were cultured in RPMI1640 media supplied with 10% fetal bovine serum before the assay. Cells were seeded into a 96-well plate at a density of 0.2 million cells per milliliter (90 μ L). A serial dilution (10 μ L, 3-fold dilution, starting a 30 μ g/mL) of anti-HER2 vc-1 was added to the appropriate wells in triplicate. After incubation at 37- $\{$ NOTE:0176 $\}$ C/5% CO₂ for 72 h, the cell viability was established using an XTT Cell Viability Assay Kit (Biotium).

HCC 1954/RAW-Dual Co-culture Assays. HCC1954 cells (ATCC) and RAW-dual cells (Invivogen) were cultured according to the manufacturer's instructions in high-glucose DMEM (Corning) supplemented with 10% FBS. Cells were pre-mixed (1:1) giving a final seeding density of 0.2 million cells per milliliter each. The cell mixture (90 μ L) was seeded into a 96-well plate and treated with 10 μ L of the appropriate test article (ADC, mAb, or PBS). The final concentration of test ADCs was 30, 6, 1.2, 0.24, and 0 μ g/mL. The naked mAb control was administered at 30 μ g/mL. The cells were cultured for 24 h at which point the supernatant was separated via centrifugation, and the activation of the NF κ B pathway was determined using the QuantiBlue assay described above.

HCC 1954/PBMC Co-culture Assays. HCC1954 cells (ATCC) and hPBMCs (iXCells Biotechnologies) were pre-mixed in RPMI1640 media (Lonzo) supplemented with 10% FBS before seeding. The final seeding densities of HCC1954 cells and hPBMCs were 0.2 million per milliliter and 2 million cells per milliliter, respectively. The cell mixture was seeded to a 96-well plate with a seeding volume of 90 μ L with 10 μ L of the appropriate test article (ADC, mAb, or PBS). The final concentration of test ADCs was 30, 6, 1.2, 0.24, and 0 μ g/mL. The naked mAb control was administered at 30 μ g/mL. The cells were cultured for 24 h, and then the supernatants were separated via centrifugation. IFN α levels in supernatant samples were quantified using a human IFN α all subtype ELISA kit (Pbl assay science, Cat# 411351).

Mouse Xenograft Efficacy Study. Human breast cancer cells (HCC1954, ATCC) were cultured in RPMI 1640 (Corning) supplied with 10% FBS (Avantar) and maintained under the manufacturer's recommended densities. 100 U mL⁻¹ penicillin–streptomycin was applied to prevent microbe contamination. Immediately before the tumor implantation, the cells were trypsinized, rinsed, and re-suspended into a suspension of 40 million cells mL⁻¹. The suspension was mixed 1:1 with Matrigel (Corning) to form the final implantation mixture, which was kept on ice for no longer than 2 h. Approximately 2 million cells (100 μ L of the mixture) were implanted subcutaneously to the right flank of 6–8 weeks old female SCID/beige mice (Charles River Labs). Tumor volume was recorded twice a week using a caliper and estimated using

Scheme 1. Reagents (i) NEt_3 , CH_2Cl_2 ; (ii) Zn , HCOONH_4 , MeOH ; (iii) $\text{C}_4\text{H}_9\text{COCl}$, NEt_3 , EtOAc ; (iv) NaOH , EtOH :water (13:2); (v) a. mCPBA , CHCl_3 ; b. $\text{NH}_4\text{OH}/\text{NH}_3$, CHCl_3 ; (vi) 20% TFA , CH_2Cl_2 ; (vii) mcValCitPABC-PNP , 2,6-Lutidine, HOBt , DMA



the following formula: $\text{length} \times \text{width}^2/2$. ADC treatment was initiated once the tumor volumes reached 50–300 mm^3 . Mice were randomly assigned to 10 different treatment groups (5 mice per group). The mice were dosed with ADC (10 or 3 mg kg^{-1}), naked anti-HER2 mAb (10 mg kg^{-1}), or DPBS via intraperitoneal injection 3 times in total with 5 day intervals. Tumor volumes and body weights were measured every 2–3 days. Mice whose tumor exceeded 1000 mm^3 , suffered from ulceration, or displayed any signs of stress during the study were euthanized based on IACUC-approved animal protocols.

RESULTS

Preparation and Evaluation of TLR-Agonist Payloads.

The efficiency of ADC payload delivery is typically limited by the number of antigens on the cell surface, thus necessitating the use of ultra-high potency payloads (typically low nanomolar or sub-nanomolar). With this in mind, we selected a panel of payloads either known or suspected to have high potency against TLR7/8 (Figure 1A). Compound 1 was specifically designed on the premise of improving the

permeability of the known TLR7/8 agonist **2**²⁸ by replacement of the primary amine (protonated at neutral pH) with an aniline (likely non-protonated at neutral pH). Indeed, $\text{logD}_{7.4}$ measurements indicated a 2 log unit increase, suggesting improved permeability (2.6 vs. 0.42)²⁹ (vide infra). Compounds 3 and 4 were designed with the hopes of picking up an additional H-bonding interaction with the receptor, as suggested by the known imidazoloquinoline agonists 5 and 7. The preparation of these compounds, as shown in Scheme 1, was adapted from the previously reported synthesis of imidazoloquinolines.²⁸ In short, an $\text{S}_{\text{N}}\text{Ar}$ reaction with 4-chloro-3-nitroquinoline (**9**) followed by a reduction gave diaminoquinoline **12a/12b** in high yield. Amidation and base-promoted dehydration gave the key tetracycle (**13a/b/c/d**). Amination ortho to the quinoline nitrogen was accomplished by N-oxidation with mCPBA , followed by a Tos-Cl -promoted rearrangement. Importantly and unexpectedly, we found that the success of this reaction necessitated the ammonium hydroxide to be added prior to the addition of TsCl . Reversing the order of addition resulted in the formation of significant

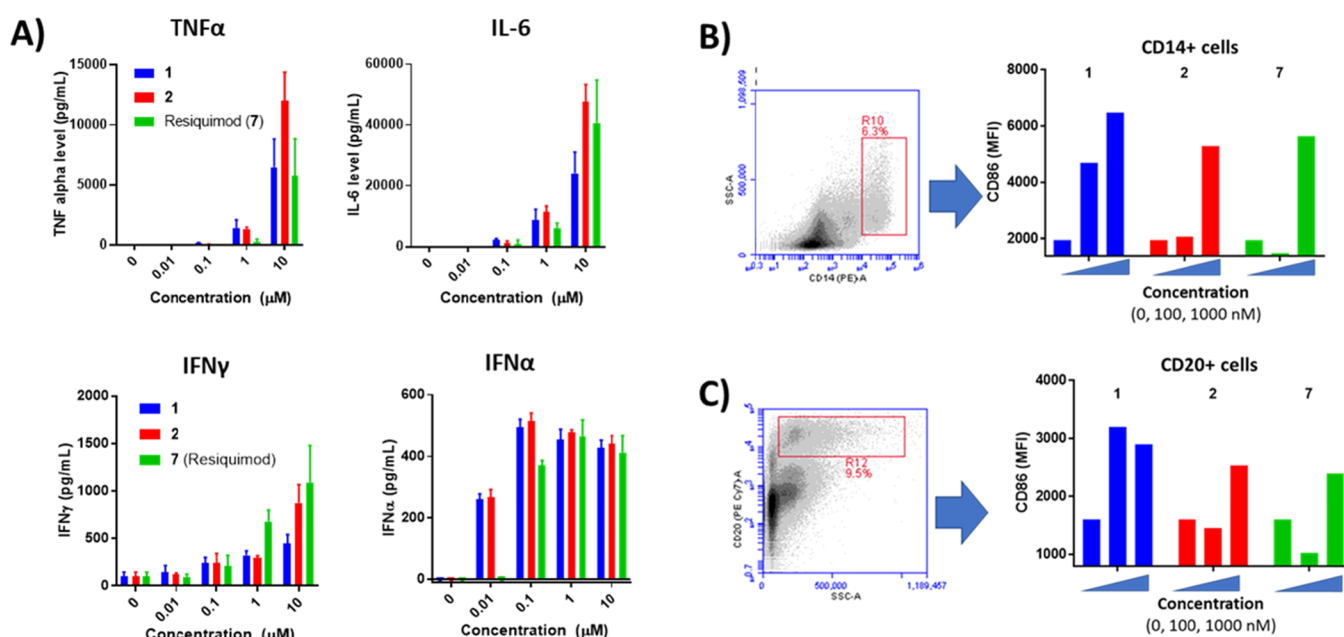


Figure 2. (A) TNF α , IL-6, IFN γ , and IFN α release was assessed following a 24 h incubation with PBMCs; (B,C) PBMCs were dosed with payloads 1, 2, or 7 (at 0, 100, or 1000 nM), and CD86 expression was assessed on CD14+ cells (B) and on CD20+ cells (C).

amounts of the 4-hydroxy imidazole[4,5-c]quinoline. The synthesis was completed by removal of the pendant Boc group with TFA.

The 8 compounds shown in Figure 1 were screened for activity using a Ramos cell line engineered with a secreted embryonic alkaline phosphatase (SEAP) reporter gene coupled to an NF κ B promoter sequence (Ramos Blue, InvivoGen). Ramos cells are a B-cell lymphoma line known to express TLR1, 2, 3, 7, and 9. Activation of the TLR receptors results in NF κ B activation and SEAP production, which can be quantified by incubating the media with a colorimetric SEAP substrate. Screening of the compounds of interest demonstrated that only compounds 1, 2, and 7 (resiquimod) significantly activated the NF κ B reporter cells during the time course of the assay (24 h) (Figure 1B). A closer evaluation of these compounds in a 72 h assay indicated that compound 1 was $\sim 5\times$ more potent than compounds 2 and 7 (Figure 1C) and that the compounds were non-cytotoxic to Ramos cells at concentrations up to 30 μ M (Figure S1). The suspected selectivity of compound 1 for TLR7 was confirmed in a hTLR7 and hTLR8 reporter assay using HEK293 cells engineered to overexpress hTLR7 or hTLR8 (InvivoGen). Once again, the cell lines contained an NF κ B response element linked to a SEAP reporter, thus allowing for the activation of the TLR pathway to be assessed using an alkaline phosphatase substrate (Figure 1D). Compound 1 activated hTLR7 at concentrations as low as 16 nM, while hTLR8 required ~ 1000 -fold higher concentration to achieve similar levels of activation, suggesting a very large selectivity for TLR7 over TLR8. The very high TLR7 selectivity was also observed in the mouse homologues of the receptors, as illustrated in Figure 1E,F. The similarities between the two receptors, combined with the similar downstream signaling pathways, have made it challenging to understand the therapeutic potential of TLR7 agonists versus TLR8 agonists. Therefore, the exquisite selectivity of compound 1 in both human and mouse will be important to understand the pharmacology of the resulting ADCs in animal studies.

While the assays in Figure 1 demonstrate the ability of compounds 1 and 2 to bind and activate the TLR7 receptor, they do not directly demonstrate an immunologically relevant response. To this end, we evaluated the ability of the TLR7 agonists to induce cytokine release in PBMCs. At low nanomolar concentrations, compounds 1 and 2 promoted significant release of IFN α but not TNF α , IL-6, or IFN γ (Figure 2A). Substantial IFN γ , IL6, and TNF α release was not observed until concentrations reached ~ 10 μ M, suggesting that this activity may be due to off-target agonism of TLR8 (see Figure 1). This is consistent with prior literature showing that selective TLR7 agonists selectively induce IFN α in PBMCs, while TLR8-selective compounds are more known for promoting the release of TNF α and IL8.¹¹ Interestingly, compounds 1 and 2 were ~ 10 -fold more potent than resiquimod (7) in terms of their induction of IFN α . Further, we also assessed whether compounds 1, 2, and 7 promoted CD86 expression on antigen-presenting cells (APCs). CD86 is an important APC co-receptor essential for the activation of naive CD28+ T-cells. Gratifyingly, we observed that CD86 was strongly induced in both CD20+ and CD14+ subpopulations of PBMCs (B-cells and myeloid cells, respectively). Interestingly, payload 1 was observed to be 5–10 $\times$ more potent than payloads 2 and 7 (Figure 2B,C).

Preparation of Tumor-Targeting and Control ADCs.

Prompted by the potent activity of compounds 1 and 2 in the functional assays described above, we proceeded to attach both to the widely known valine-citrulline p-aminobenzyl carbamate (ValCitPABC) linker, giving mc-ValCitPAB-1 (or "vc-1") and mcValCitPABC-2 (or "vc-2") (Scheme 1). Structures were confirmed by NMR and LCMS, as described in the Experimental Section. The linker-attachment site was further demonstrated by LC–MS/MS analysis showing clear fragmentation at the benzylic position with loss of the linker (Figure S2). Resiquimod (7) was attached to the cleavable mcValCitPABC linker as a comparator, this time via the 4-amino functionality of the imidazoloquinoline core (mcValCitPABC-7) (Scheme 1). We attached these linker payloads to

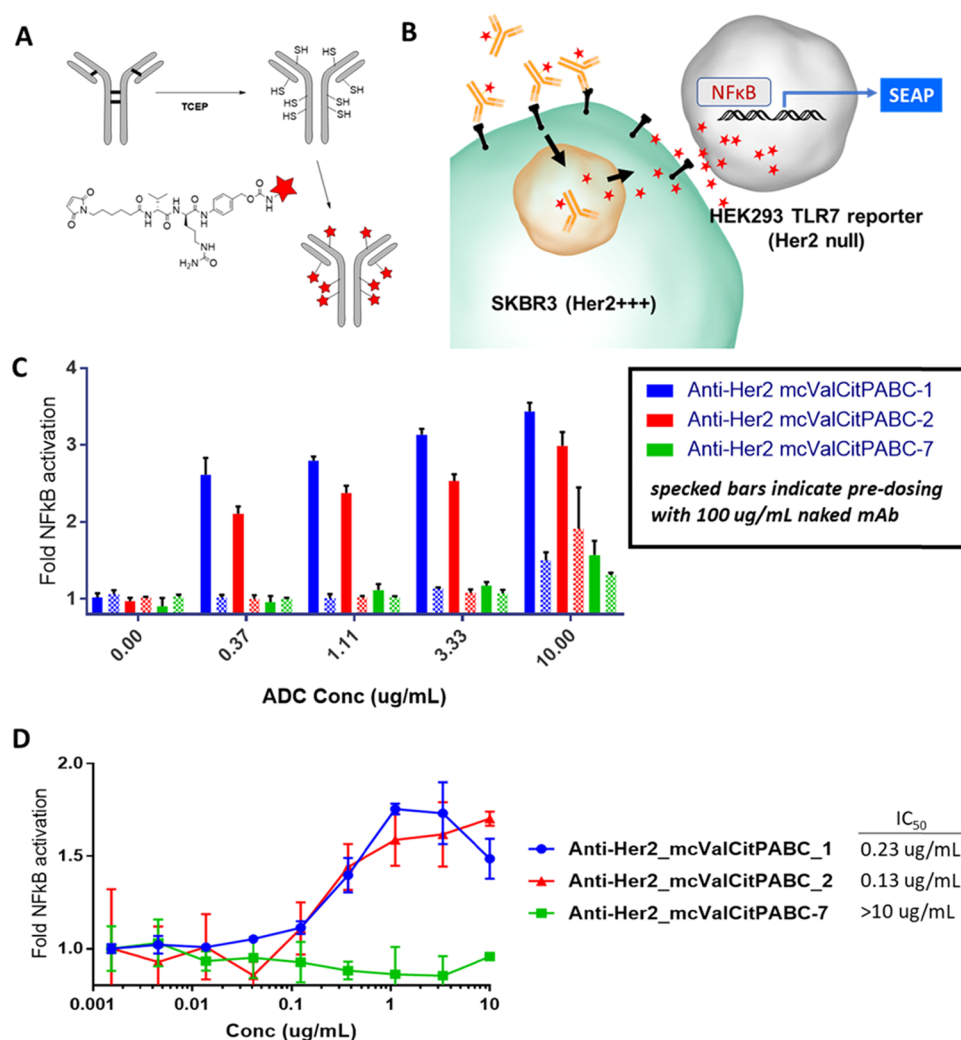


Figure 3. Media transfer experiments demonstrating release of payload by LCMS (A) and activation of a HEK293 TLR7 reporter cell line (B–D).

trastuzumab, a HER2-targeting antibody that is widely used for the treatment of breast cancer. We elected to prepare tumor-targeting ADCs (anti-HER2) and non-targeted control ADCs (anti-CD20 or anti-RSV) with a DAR of 6–8 in order to maximize the in vitro potency during the discovery process. While a high DAR (>4) is often associated with poor PK, it is also well established that a high DAR results in superstoichiometric enhancements in potency, thus facilitating ADC discovery efforts.³⁰ The conjugation was accomplished using 6–10 equiv of TCEP as a reducing agent to completely cleave the endogenous inter-chain disulfides (Figure 3A). After the removal of excess TCEP, the antibody was incubated with the appropriate maleimide linker-payload for 2–16 h. The final loading was typically between 6 and 8 drugs per antibody, and most samples exhibited <5% aggregate content (see Table 1). Note that all ADCs were prepared using antibodies that had been deglycosylated prior to conjugation in order to minimize any complications arising from uptake by Fc-γ receptors on lymphocytes.

Anti-HER2 Immune-Stimulating ADCs Release Payload into Cell Media and Indirectly Activate TLR7-Expressing Cells in the Presence of HER2+ Breast Cancer Cells. Our overall objective was to design an ADC delivery system that releases a highly permeable TLR7 agonist in tumor tissue which can subsequently escape from the

targeted cells and have a pharmacological impact on nearby cells. This so-called “bystander effect” is commonly observed in various ADCs that deliver cell-permeable payloads.³¹ With this in mind, we designed an LCMS assay in which SK-BR-3 breast cancer cells (HER2+++), HEK293 TLR7 reporter (Her2 null), and HEK293 TLR7 reporter (Her2 null) cells were incubated with 10 μg/mL of either a targeted ADC (Anti-HER2 mcValCitPABC-2) or an analogous null-targeted ADC (Anti-CD20 mcValCitPABC-2). Aliquots of growth media were removed at various times and diluted with acetonitrile to precipitate media proteins. The amount of payload was assessed by LCMS/MS (360.2 → 93.0) (Figure S3). We observed that the tumor-targeted ADC (anti-HER2) released a significant payload into the media by 4 h and levels rapidly increased over 24 h. The corresponding null-targeted ADC (anti-CD20) released only negligible amounts of payload during the same period. Pre-dosing with naked anti-HER2 suppressed the amount of payload released at 4 h but not at later time points (perhaps due to the de novo expression of HER2). An analogous experiment was performed, in which SK-BR-3 cells were incubated with various concentrations of a targeted and non-targeted ADC for 48 h, followed by transfer of the conditioned media into a HEK293 mTLR7 reporter cell line, as described above. Consistent with the LCMS results, the targeted (anti-HER2) ADC elicited a strong NFκB response, while the non-targeted ADC did not (Figure S3). Interestingly, co-dosing the targeted ADC with 10× naked anti-HER2

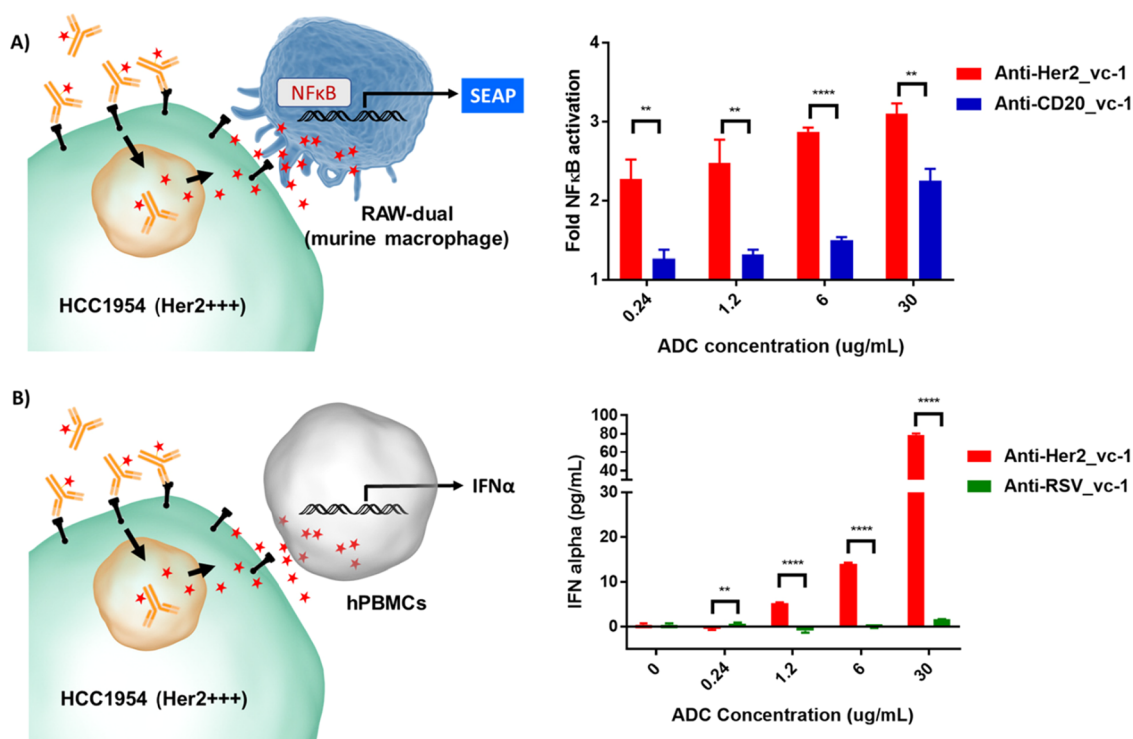


Figure 4. (A) HCC1954 cells were co-cultured with the mouse macrophage reporter cell line RAW-dual. Targeted ADC treatment (anti-HER2) resulted in activation of the NFκB pathway, while non-targeted ADC treatment (anti-CD20) did not. (B) HCC1954 cells were co-cultured with hPBMCs in the presence of a targeted (anti-HER2) or non-targeted (anti-RSV) ADC. After 24 h, the medium was assessed for IFNα by ELISA.

significantly suppressed the bystander activity. However, the suppression did not bring the NFκB activity of the reporter cells back to baseline, suggesting incomplete blockage of ADC uptake, which is consistent with the LCMS results.

Anti-HER2_vc-1 and Anti-HER2_vc-2 Exhibit Significantly Greater Bystander Activity Than Anti-HER2_vc-7.

It is widely known that the physiochemical properties of the released species drive the bystander activity of ADCs.³¹ Therefore, ADCs delivering the three payloads of interest (1, 2, and 7) were evaluated in an SK-BR-3 media-transfer assay using the mTLR7 reporter cell line described above (Figure 3B,C). Interestingly, we observed that ADCs conjugated to payloads 1 and 2 exhibited significant indirect activation of the mTLR7 reporter cells while those conjugated to payload 7 did not. The reason behind this may be multifaceted, including factors such as payload potency, rate of linker release, and rate of lysosomal escape. Nonetheless, it is interesting to note that anti-HER2 mcValCitPABC-7 did not elicit any significant activation of the reporter cells, even at concentrations as high as 10 μg/mL, while the analogous ADCs with payloads 1 and 2 were active at concentrations as low as 0.37 μg/mL (Figure 3C). Importantly, the addition of 100 μg/mL of naked antibody (anti-HER2) to the assay effectively suppressed the activity of the ADCs, strongly supporting the antigen-mediated uptake mechanism. A further dose-titration experiment was performed in order to establish approximate IC₅₀ values for the ADCs (Figure 3D). Both anti-HER2 mcValCitPABC-1 and anti-HER2 mcValCitPABC-2 exhibited an IC₅₀ of ~0.1 to 0.2 μg/mL, while anti-HER2 mcValCitPABC-7 showed little or no activity at concentrations up to 10 μg/mL. The inactivity of this ADC in the bystander assays is likely derived from a combination of reduced payload potency (Figures 1C and 2A) and reduced permeability. The cLogP of compound 7 (1.84) is

considerably lower than those of compounds 1 and 2 (3.92 and 4.10, respectively), suggesting reduced permeability. However, the measured shake-flask LogD_{7.4} of 1, 2, and 7 does not support this hypothesis (2.61, 0.42, and 1.14, respectively) (Figure S4). However, it is important to point out that the process of lysosomal escape is poorly understood, and there is much speculation that lysosomal membrane transporters may play a significant role in payload escape, in which case payload permeability is only part of the equation. In spite of the nuances of payload permeability, it is clear that payloads 1 and 2 are able to escape from antigen-expressing tumor cells and have an impact on nearby TLR7-expressing cells. Similar to the data in Figures 2B,C, and 1C, compound 1 was slightly (but consistently) more potent than compound 2. Thus, from this point onward, we limited our focus to ADCs derived from compound 1.

Anti-HER2_vc-1 Activates Tumor-Associated Mouse Macrophages and hPBMCs When Co-cultured with HER2+++ HCC1954 Cells. Our previous assays (above) demonstrated that anti-HER2 ADCs were internalized and processed by SKBR3 cells, resulting in the release of the TLR7 agonist into the cell media. However, it was not clear if the concentration of the released payload would be sufficient to elicit a functional response in tumor-associated lymphocytes. Around this time, a paper by Ackerman was published, demonstrating that a class of non-cleavable TLR7 agonist ADCs were able to cause tumor regression in an HCC1954 (HER2+++) xenograft study in SCID beige mice.²¹ Building on this work, we evaluated anti-HER2_vc-1 in a co-culture assay with HCC1954 cells and RAW-dual cells (Invivogen), a mouse macrophage cell line engineered with an NFκB and IRF reporter system. As shown in Figure 4A, anti-HER2_vc-1 potently induced NFκB activation in mouse macrophages,

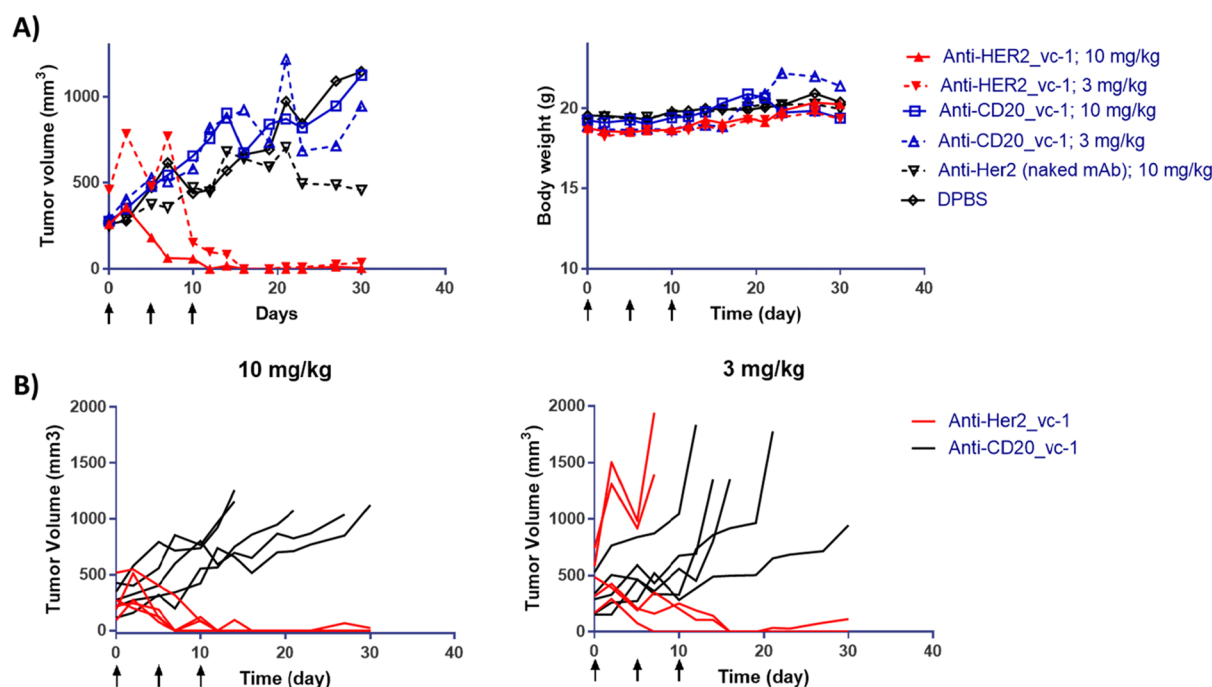


Figure 5. SCID beige mice were implanted with HCC1954 cells and were randomized until the tumor volume reached ~ 200 to 400 mm^3 , at which time a targeted (**anti-HER2_vc-1**) or non-targeted (**anti-CD20_vc-1**) ADC was administered by IP injection ($Q4d \times 3$). Tumor volume and weight were recorded every 2–3 days.

while an isotype control conjugate (**anti-CD20_vc-1**) did not. (Note that both RAW-dual cells and HCC1954 cells do not express CD20.) The non-targeted control ADC activated the mouse macrophages only at high concentrations ($30 \mu\text{g/mL}$), presumably due to non-specific uptake by micropinocytosis. Note that both ADCs were prepared from deglycosylated antibodies in order to minimize uptake by Fc- λ receptors on lymphocytes.

While these results suggest that the tumor-targeted **anti-HER2_vc-1** will activate tumor-associated mouse macrophages, we wanted to perform an analogous experiment using human primary cells in order to increase the confidence that this approach has a clinical value. Unfortunately, human macrophage cell lines, such as THP1, are known to express only low levels of TLR7.³² We confirmed this finding, showing that low micromolar levels of compounds **1** and **2** were required to induce TNF α in THP1-derived macrophages (Figure S5). However, our prior results showing that low nanomolar concentrations of **1** potentially induced IFN α release in PBMCs prompted us to co-culture HCC1954 cells with hPBMCs in the presence of a targeted (**anti-HER2_vc-1**) and non-targeted (**anti-RSV_vc-1**) TLR7 ADC (Figure 4B). Note that the previously mentioned **anti-CD20_vc-1** would be an unsuitable negative control in this case due to the presence of CD20 on primary human B-cells. Anti-RSV (pavilizumab) is an IgG1 antibody that binds to the coated protein of the respiratory syncytial virus (RSV). This antibody is frequently employed as a non-binding “isotype control” in ADC studies.³³ Gratifyingly, treatment with **anti-HER2_vc-1** resulted in the release of IFN α from PBMCs, while treatment with the isotype control did not, even at high concentrations of ADC (Figure 4B). Once again, both the isotype control and the targeted ADC were prepared from deglycosylated antibodies devoid of binding affinity to most Fc- λ receptors.²⁷ The potent activity observed for these deglycosylated ADCs in the two co-culture

assays is in direct contrast to the report by Ackerman, showing that glycosylation is essential for immunostimulatory activity.²¹ We attribute the discrepancy between these studies to the fundamentally different nature of the linker employed (cleavable vs. non-cleavable), as will be further discussed below.

Anti-HER2_vc-1 Induces Tumor Regression in an HCC1954 Mouse Xenograft Model. Building on successful co-culture results, we implanted HCC1954 cells into SCID beige mice. After the tumors reached ~ 200 to 400 mm^3 , the mice were treated with either the targeted (**anti-HER2_vc-1**) or non-targeted (**anti-CD20_vc-1**) at either 3 mg per kg (mpk) or 10 mpk (IP, $Q5dx3$). Vehicle-treated and naked mAb-treated control groups were included for comparison. Both cohorts of HER2-targeted ADCs exhibited rapid tumor regression, resulting in complete disappearance of tumor by 15 days post-treatment (Figure 5A). Note that two animals in the 3mpk group failed to respond to treatment, likely because the starting tumor volume exceeded 500 mm^3 (Figure 5B). No tumor regression was observed in animals treated with the control ADC (**anti-CD20_vc-1**) or the naked mAb (**anti-HER2**). It should be noted that the anti-CD20 (rituximab) used for the isotype ADC does not bind to mouse CD20 and thus was treated as a non-binding control for the purpose of this study.³⁴ No significant changes in weight or behavior were noted for either the targeted or non-targeted ADC, as compared to the vehicle control (Figure 5A). Importantly, **anti-HER2_vc-1** was not toxic to either HCC1954 or SKBR3 cells, indicating that the tumor regression is due to indirect activation of tumor-associated myeloid cells rather than to direct tumor toxicity (Figure S6).

DISCUSSION

The vast majority of ADCs programs for the treatment of cancer have focused on the delivery of highly cytotoxic agents

to tumor tissue. While this has been a highly fruitful endeavor, the advent of checkpoint inhibitors and other immunoncology therapeutics have revolutionized treatment regimens and have shifted the focus of drug-discovery efforts toward agents that can selectively and safely activate innate immune pathways. TLRs form a key link between the innate and adaptive immune systems and therefore have been the target of considerable interest for pharmaceutical intervention. TLRs specifically respond to DAMPs that often arise from bacterial, viral, and fungal infections. Downstream signaling from TLRs leads to NF κ B and IRF activation which, in turn, leads to the production of pro-inflammatory cytokines and type-1 interferons that facilitate T-cell recruitment and activation. As such, low-dose TLR agonists (i.e., TLR7 and TLR8) have been widely used as vaccine adjuvants but have received only modest attention as cancer therapeutics due to the risk of systemic toxicity.¹⁶ With this in mind, we undertook the design of ADCs that selectively deliver a potent TLR7 agonist to the antigen-expressing tissue.

Unlike others in the field who rely on the Fc- γ -mediated uptake of non-cleavable linkers, our focus was on cleavable linkers attached to cell-permeable payloads that may activate tumor-associated lymphocytes in a classical “bystander” fashion.²¹ Membrane permeability, therefore, became a key criterion for the selection of payloads. After screening a small set of TLR7 agonists, we settled on imidazoloquinolines **1** and **2** as a highly potent and selective TLR7 modulators with built-in amine handles for facile linker attachment. We were particularly interested in compound **1** due to the reduced basicity of the pendant amine that imparted an increase in LogD, a key parameter known to be important for membrane permeability. Both compounds potently activate the NF κ B pathway in hTLR7 (but not hTLR8) reporter cells, mTLR7 (but not mTLR8) reporter cells, and Ramos-blue cells (a Burkitt cell lymphoma) (Figure 1). In all cases, the activity of these compounds was equal or better than the well-studied comparator compound, resiquimod (**7**). Furthermore, at low nanomolar concentrations, these compounds induce the release of IFN α (but not TNF α or IL6) in hPBMCs and induce the expression of CD86 (a T-cell co-receptor) in various antigen-expressing cells (Figure 2). Taken together, this data demonstrates that compounds **1** and **2** potently activate TLR-signaling pathways that make them amenable for use as an ADC payload. Attachment to the well-studied anti-HER2 (trastuzumab) backbone via conjugations to endogenous cysteine residues resulted in an ADC that rapidly released payload into the media and activated bystander cells in an SKBR3 $\rightarrow$ TLR7 reporter media transfer assay (Figures 3 and S3). Importantly, this activity was suppressed by co-treatment with excess naked antibody, demonstrating the antigen-mediated uptake mechanism of the ADCs. Taken together, these experiments strongly support the “bystander” mechanism of action for the tumor-targeted immune-stimulating ADCs.

Having established the basic proof-of-mechanism, we proceeded to evaluate the ability of the tumor-targeting ADC (anti-HER2_vc-1) to promote macrophage activation and IFN α release in a co-culture assay. Incubation of ADCs in the presence of both HCC1954 (HER2+++) and mouse macrophages confirmed that the macrophage NF κ B pathway was only activated in the presence of the tumor-targeted ADC (Figure 4A). Likewise, in a HCC1954 + hPBMC co-culture experiment, IFN α release was observed with the tumor-targeted (anti-HER2) but not the non-targeted (anti-RSV)

ADC. Finally, the efficacy of anti-HER2_vc-1 was established in an HCC1954 xenograft study using SCID beige mice. SCID beige mice lack functional B-cells, T-cells, and NK cells but produce normal macrophages, neutrophils, basophils, and eosinophils. Based on the potent activation of RAW-dual cells observed in the co-culture assay (Figure 4A), we hypothesize that the anti-tumor activity of this ADC is derived from its ability to potently and selectively activate TAMs which, in turn, phagocytose the tumor tissue.³⁵ While the SCID beige mice are devoid of an adaptive immune system, engulfment by TAMs (or other APCs) is a critical first step to activation of a sustained T-cell response. The upregulation of CD86 in human APCs that is induced in these TLR7 agonists (Figure 2) is viewed as a key feature that may drive the generation of memory T-cells needed for tumor immunogenicity.

Systemic exposure to TLR agonists is widely known to elicit dose-limiting side effects that prevent their use in the treatment of cancer. Multiple groups have attempted to address this limitation by using tumor-targeting technologies such as nanoparticle delivery and intratumoral injections.^{13,17,18} However, there have only been two reports to date describing the use of ADC technology for the delivery of TLR agonists, both of which use non-cleavable linkers and are unlikely to exhibit any meaningful bystander activity in tumor-associated lymphocytes.^{17,21} Indeed, the mechanism postulated by Ackerman requires opsonization of antigen-expressing tumor cells which, in turn, are phagocytosed by tumor-associated APCs. Importantly, Ackerman showed that this requires Fc- γ -mediated uptake. In their hands, ablation of the Fc- γ -binding affinity (by using deglycosylated antibodies) resulted in ADCs with no meaningful anti-tumor activity. In contrast, by moving to a cleavable linker attached to a cell-permeable payload, we were able to design potent immune-stimulating ADCs that do not rely on antibody glycosylation. Indeed, all of the experiments described herein employed ADCs that were devoid of glycosylation. This is particularly important because immune-stimulating ADCs have a very significant risk of immunogenicity through the development of anti-drug antibodies (ADA). Indeed, a recent phase I study from Novartis demonstrated that 14 out of 14 patients dosed with an immune-stimulatory ADC developed a measurable ADA response.³⁶ Elimination of Fc- γ -mediated uptake into APCs is one mechanism that is being explored for attenuating ADA responses of biotherapeutics.³⁷ Removal of the antibody glycosyl group is a well-known strategy for eliminating most Fc- γ -binding interactions.²⁷ Thus, the work described herein paves the way for tumor-specific delivery of TLR7 agonists using ADCs with antibody frameworks that are devoid of Fc- γ -binding affinity, thus making them less likely to induce a deleterious ADA response. Future work will focus on optimizing the potency, stability, and PK of the ADCs in order to facilitate in vivo efficacy studies in syngeneic mouse models. We believe that this ADC technology for the delivery of potent and selective TLR7 agonists will prove to be a valuable tool in the growing repertoire of tumor-targeted immune-modulating agents.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.molpharmaceut.2c00392>.

Preparation and evaluation of MMAE ADCs as well as the NF κ B activation by non-targeted ADCs and linker-payloads (PDF)

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Notes

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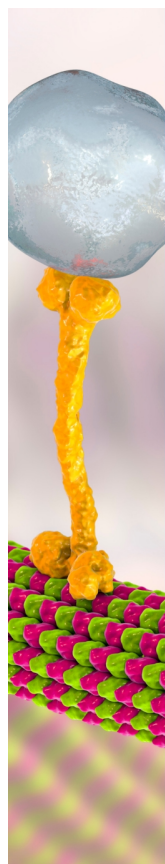
ABBREVIATIONS

ACN, acetonitrile; ADC, antibody-drug conjugate; DAR, drug-antibody ratio; DMA, dimethylacetamide; DCM, dichloromethane; DMSO, dimethylsulfoxide; DPBS/PBS, phosphate-buffered saline; ELISA, enzyme-linked immunosorbent assay; FBS, fetal bovine serum; HATU, hexafluorophosphate azabenzotriazole tetramethyl uronium; HOBt, hydroxybenzotriazole; IFN, interferon; IL-6, interleukin-6; LCMS, liquid chromatography/mass spectroscopy; MFI, mean fluorescence intensity; PBMC, peripheral blood mononucleocytes; mAb, monoclonal antibody; mCPBA, meta-chloroperoxybenzoic acid; MMAE, monomethyl auristatin E; NSCLC, non-small cell lung cancer; pDCs, plasmacytoid dendritic cells; SEC, size-exclusion chromatography; SEAP, secreted embryonic alkaline phosphatase; TAM, tumor-associated macrophage; TNF α , tumor necrosis factor alpha; TLR, toll-like receptor; TFA, trifluoroacetic acid; TCEP, tris(2-carboxyethyl)phosphine

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